SAEON Agile Logbook

Name: Angelo Adams

Student Number: 230450431

Field	Record
Student number	To be completed by student
Role	Frontend Developer and Designer
Project	SAEON Real-Time Rainfall Monitoring System
Reporting period	28 July to 23 October 2026
Working days	Tuesday to Friday
Revision date	27 October 2026

Purpose and project configuration

This individual logbook records progress planned next steps and impediments against the project's 13-week schedule. Weekly headings identify the planned work area; the entries record individual contributions, including supporting work outside a person's main role.

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is a database platform.

The agreed architecture retains SIM7000G cellular communication and HiveMQ Cloud MQTT, with a backend ingestion service writing to Firebase. Final deployment and field validation remain open.

Record status

The daily entries are reconstructed working records and require confirmation by the named student. They are not verified in attendance or proof that a task was completed. Impediments describe possible risks unless a dated incident is confirmed. Correct any inaccurate activity before signing the logbook.

This revision confirms the technology choices; it does not verify past daily activities or create missing test results. Proposed current follow-up actions appear after the dated entries.

Daily questions

- What did I complete since the last standup?
- What am I planning to complete between now and the next standup?
- What are my impediments (or risks or problems)?

The tables use shortened labels for these same three questions. Mondays are outside the supplied reporting schedule; no Monday attendance or activity is inferred. Historical stand-up meetings are not implied by daily entries.

Week 1 28 July 2026 to 31 July 2026

Planned work area: Project Initiation & System Architecture
PB01 | US01, US02, US03, US04, US13

Tuesday 28 July 2026

Completed since the last stand-up	I reviewed and documented dashboard user needs.
Plan before the next stand-up	I will continue with sensor display requirements and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Wednesday 29 July 2026

Completed since the last stand-up	I worked on sensor display requirements.
Plan before the next stand-up	I will continue with frontend development setup and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Thursday 30 July 2026

Completed since the last stand-up	I checked and updated frontend development setup.
Plan before the next stand-up	I will continue with dashboard data-flow sketch and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Friday 31 July 2026

Completed since the last stand-up	I organised the evidence for dashboard data-flow sketch.
Plan before the next stand-up	I will continue with serial-to-display mapping and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Week 2 4 August 2026 to 7 August 2026

Planned work area: Core Firmware & Sensor Interfacing
PB02 | US01, US02, US11

Tuesday 4 August 2026

Completed since the last stand-up	I reviewed and documented serial-to-display mapping.
Plan before the next stand-up	I will continue with temperature labels and error states and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Wednesday 5 August 2026

Completed since the last stand-up	I worked on temperature labels and error states.
Plan before the next stand-up	I will continue with rainfall units and tip displays and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Thursday 6 August 2026

Completed since the last stand-up	I checked and updated rainfall units and tip displays.
Plan before the next stand-up	I will continue with low-power station display needs and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Friday 7 August 2026

Completed since the last stand-up	I organised the evidence for low-power station display needs.
Plan before the next stand-up	I will continue with broker data requirements and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Week 3 11 August 2026 to 14 August 2026

Planned work area: HiveMQ Cloud Broker Setup
PB03 | US03, US04, US09

Tuesday 11 August 2026

Completed since the last stand-up	I reviewed and documented broker data requirements.
Plan before the next stand-up	I will continue with dashboard topic subscription design and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Wednesday 12 August 2026

Completed since the last stand-up	I worked on dashboard topic subscription design.
Plan before the next stand-up	I will continue with sample MQTT display checks and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Thursday 13 August 2026

Completed since the last stand-up	I checked and updated sample MQTT display checks.
Plan before the next stand-up	I will continue with Firebase dashboard access design and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Friday 14 August 2026

Completed since the last stand-up	I organised the evidence for Firebase dashboard access design.
Plan before the next stand-up	I will continue with cellular status display design and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Week 4 18 August 2026 to 21 August 2026

Planned work area: Cellular Communication (SIM7000G)
PB04 | US03, US04

Tuesday 18 August 2026

Completed since the last stand-up	I reviewed and documented cellular status display design.
Plan before the next stand-up	I will continue with network-state wording and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Wednesday 19 August 2026

Completed since the last stand-up	I worked on network-state wording.
Plan before the next stand-up	I will continue with live cellular reading display checks and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Thursday 20 August 2026

Completed since the last stand-up	I checked and updated live cellular reading display checks.
Plan before the next stand-up	I will continue with offline and reconnect interface checks and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Friday 21 August 2026

Completed since the last stand-up	I organised the evidence for offline and reconnect interface checks.
Plan before the next stand-up	I will continue with application skeleton and routing and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Week 5 25 August 2026 to 28 August 2026

Planned work area: SAEON Dashboard Base Application
PB05 | US05, US06, US09

Tuesday 25 August 2026

Completed since the last stand-up	I reviewed and documented application skeleton and routing.
Plan before the next stand-up	I will continue with real-time visualisation and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Wednesday 26 August 2026

Completed since the last stand-up	I worked on real-time visualisation.
Plan before the next stand-up	I will continue with historical data and filtering and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Thursday 27 August 2026

Completed since the last stand-up	I checked and updated historical data and filtering.
Plan before the next stand-up	I will continue with cloud-command interface preparation and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Friday 28 August 2026

Completed since the last stand-up	I organised the evidence for cloud-command interface preparation.
Plan before the next stand-up	I will continue with battery-status display design and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Week 6 1 September 2026 to 4 September 2026

Planned work area: Power Supply & Solar Integration
PB06 | US11

Entry status: reconstructed; student confirmation required.

Tuesday 1 September 2026

Completed since the last stand-up	I reviewed and documented battery-status display design.
Plan before the next stand-up	I will continue with solar-status information needs and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Wednesday 2 September 2026

Completed since the last stand-up	I worked on solar-status information needs.
Plan before the next stand-up	I will continue with power-source display notes and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Thursday 3 September 2026

Completed since the last stand-up	I checked and updated power-source display notes.
Plan before the next stand-up	I will continue with power-related interface review and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Friday 4 September 2026

Completed since the last stand-up	I organised the evidence for power-related interface review.
Plan before the next stand-up	I will continue with station-health diagnostics component and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Week 7 8 September 2026 to 11 September 2026

Planned work area: Advanced UI & Health Diagnostics
PB07 | US07, US08, US10

Entry status: reconstructed; student confirmation required.

Tuesday 8 September 2026

Completed since the last stand-up	I reviewed and documented station-health diagnostics component.
Plan before the next stand-up	I will continue with CSV export interface support and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Wednesday 9 September 2026

Completed since the last stand-up	I worked on CSV export interface support.
Plan before the next stand-up	I will continue with mobile and tablet layouts and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Thursday 10 September 2026

Completed since the last stand-up	I checked and updated mobile and tablet layouts.
Plan before the next stand-up	I will continue with dashboard pipeline checks and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Friday 11 September 2026

Completed since the last stand-up	I organised the evidence for dashboard pipeline checks.
Plan before the next stand-up	I will continue with station assembly diagrams for display and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Week 8 15 September 2026 to 18 September 2026

Planned work area: Hardware Finalization & PCB Soldering
PB08 | US12

Entry status: reconstructed; student confirmation required.

Tuesday 15 September 2026

Completed since the last stand-up	I reviewed and documented station assembly diagrams for display.
Plan before the next stand-up	I will continue with hardware image organisation and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Wednesday 16 September 2026

Completed since the last stand-up	I worked on hardware image organisation.
Plan before the next stand-up	I will continue with enclosure images for documentation and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Thursday 17 September 2026

Completed since the last stand-up	I checked and updated enclosure images for documentation.
Plan before the next stand-up	I will continue with station imagery and labels and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Friday 18 September 2026

Completed since the last stand-up	I organised the evidence for station imagery and labels.
Plan before the next stand-up	I will continue with power-diagnostic display checks and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Week 9 22 September 2026 to 25 September 2026

Planned work area: Pre-Deployment Auditing & Calibration
PB09 | US11, US12

Entry status: reconstructed; student confirmation required.

Tuesday 22 September 2026

Completed since the last stand-up	I reviewed and documented power-diagnostic display checks.
Plan before the next stand-up	I will continue with charging-status labels and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Wednesday 23 September 2026

Completed since the last stand-up	I worked on charging-status labels.
Plan before the next stand-up	I will continue with deployment display preparations and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Thursday 24 September 2026

Completed since the last stand-up	I prepared notes on deployment display preparations. The field work remains in progress.
Plan before the next stand-up	I will continue with field-calibration display preparations and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Friday 25 September 2026

Completed since the last stand-up	I prepared notes on field-calibration display preparations. The field work remains in progress.
Plan before the next stand-up	I will continue with field-monitoring dashboard preparation and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Week 10 29 September 2026 to 2 October 2026

Planned work area: 48-Hour Field Stress Test
PB10 | US01, US02, US03, US11, US12

Entry status: reconstructed; student confirmation required.

Tuesday 29 September 2026

Completed since the last stand-up	I prepared notes on field-monitoring dashboard preparation. The field work remains in progress.
Plan before the next stand-up	I will continue with available field-data displays and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Wednesday 30 September 2026

Completed since the last stand-up	I prepared notes on available field-data displays. The field work remains in progress.
Plan before the next stand-up	I will continue with reference-measurement display support and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Thursday 1 October 2026

Completed since the last stand-up	I prepared notes on reference-measurement display support. The field work remains in progress.
Plan before the next stand-up	I will continue with provisional field-chart review and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Friday 2 October 2026

Completed since the last stand-up	I prepared notes on provisional field-chart review. The field work remains in progress.
Plan before the next stand-up	I will continue with dashboard presentation outline and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Week 11 6 October 2026 to 9 October 2026

Planned work area: Presentation Formatting & Content
PB11 | US13

Entry status: reconstructed; student confirmation required.

Tuesday 6 October 2026

Completed since the last stand-up	I reviewed and documented dashboard presentation outline.
Plan before the next stand-up	I will continue with interface architecture diagrams and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Wednesday 7 October 2026

Completed since the last stand-up	I worked on interface architecture diagrams.
Plan before the next stand-up	I will continue with desktop and phone screenshots and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Thursday 8 October 2026

Completed since the last stand-up	I checked and updated desktop and phone screenshots.
Plan before the next stand-up	I will continue with frontend setup documentation and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Friday 9 October 2026

Completed since the last stand-up	I organised the evidence for frontend setup documentation.
Plan before the next stand-up	I will continue with final dashboard slide review and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Week 12 13 October 2026 to 16 October 2026

Planned work area: Rehearsal & Submission Prep
PB12 | US13

Entry status: reconstructed; student confirmation required.

Tuesday 13 October 2026

Completed since the last stand-up	I reviewed and documented final dashboard slide review.
Plan before the next stand-up	I will continue with dashboard demonstration rehearsal and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Wednesday 14 October 2026

Completed since the last stand-up	I worked on dashboard demonstration rehearsal.
Plan before the next stand-up	I will continue with timed demonstration refinements and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Thursday 15 October 2026

Completed since the last stand-up	I checked and updated timed demonstration refinements.
Plan before the next stand-up	I will continue with frontend evidence package and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Friday 16 October 2026

Completed since the last stand-up	I organised the evidence for frontend evidence package.
Plan before the next stand-up	I will continue with prototype interface review and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Week 13 20 October 2026 to 23 October 2026

Planned work area: Project Buffer & Final Review
PB13 | US12, US13

Entry status: reconstructed; student confirmation required.

Tuesday 20 October 2026

Completed since the last stand-up	I reviewed and documented prototype interface review.
Plan before the next stand-up	I will continue with final screenshot checks and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Wednesday 21 October 2026

Completed since the last stand-up	I worked on final screenshot checks.
Plan before the next stand-up	I will continue with open field-display limitations and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Thursday 22 October 2026

Completed since the last stand-up	I checked and updated open field-display limitations.
Plan before the next stand-up	I will continue with dashboard handover review and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Friday 23 October 2026

Completed since the last stand-up	I organised the evidence for dashboard handover review.
Plan before the next stand-up	I will continue with the remaining deployment and field-validation actions and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.